

Embedded System
Serial port communication with Blink LEDs
Project Proposal(Assignment)



Submitted by:

Huzaifa Shahbaz
21-CE-009

Aqsa Shah
21-CE-039

Submitted To:

Sir Raza Ali Shah
Department of Computer Engineering,
HITEC University, Taxila

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ABSTRACT:

The "Serial Port Communication with Blink LEDs" project has designed a serial communication that can control the LED states through a microcontroller. This, in fact, connects a computer with a microcontroller so that commands will be sent to and processed by the microcontroller. The microcontroller has to decode these commands then toggle and blink LEDs in different patterns. Thus, this project combines embedded systems and serial protocols in real-time control applications.

INTRODUCTION:

Serial communication is an important mode of data transfer between two devices. This is widely used in embedded systems for communication that is efficient and reliable. Hence, this project titled 'Serial Port Communication with Blink LEDs' deals with the serial communication protocols that help control the speed and state of LEDs only. Here, LED circuits are connected between the public computer and the microcontroller via a serial port and commands from the computer for on, off and blink states for model LEDs can be sent via a computer to the microcontroller. This brings together an interaction between hardware and software, showcasing how microcontrollers interpret incoming data through serial interfaces and perform their programmed actions. This is a pretty simple application, yet very effective, demonstrating some important elements of communication protocols and an embedded system control.

OBJECTIVES:

1. Learn the concepts of serial communication and implement it into embedded systems.
2. Create a system for reliable communication between a computer and a microcontroller using a serial port connection.
3. Control the LED by decoding the serial commands sent from the computer to the microcontroller.
4. Execute different LED blinking patterns as a result of commands received from the microcontroller.
5. Exhibit the application of serial communication in practical hardware control in real time.

COMPONENTS USED:

- 1: ARDUINO
- 2: MATLAB

PROBLEM STATEMENT:

Usually, a device such as LEDs is operated by direct human interface or hardcoded logic, which makes it harder to operate or control dynamically. Therefore, easy, efficient mechanism required to control LEDs remotely and possibly run various patterns in real time. Serial communication is seen to provide such a way out by creating seamless communication between computer and microcontroller so that hardware control commands could be executed accordingly. This mobile project handles the major task of serial communication of LED control, which may have various means of software and embedded system integration.

METHODOLOGY:

This methodology develops a serial interface between a computer and an Arduino to define a protocol for controlling LED actions. It also teaches the microcontroller how to decode commands and uses terminal software for testing and debugging. Improvements are made by optimizations for data transfer through error handling and feedback mechanism, followed by demonstrating variable control over LEDs through real-time serial communication. This is ensured by integrating all components for full demonstration of dynamic control over LEDs through real-time serial communication.

CONCLUSION:

In conclusion, "Serial Port Communication with Blink LEDs" is a project that proves working with serial communications for any hardware control, in real time. With a successful setup of the reliable interface connecting through a microcontroller with a computer under simple commands, the LED states can be altered dynamically. This project is the best example to show how well software and hardware can merge together. It also portrays very practical applications of serial protocols in embedded systems. Indeed, this is the basis for more complex systems that demand real-time communication and control, confirming their great role in the versatility and efficiency of serial communication for modern electronics.